

Learning English to Chinese Character: Calligraphic Art Production based on Transformer

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ABSTRACT

We propose a transformer-based model to learn Square Word Calligraphy to write English words in the format of a square that resembles Chinese characters. To achieve this task, we compose a dataset by collecting the calligraphic characters created by artist Xu Bing, and labeling the position of each alphabet in the characters. Taking the input of English alphabets, we introduce a modified transformer-based model to learn the position relationship between each alphabet and predict the transformation parameters for each part to reassemble them as a Chinese character. We show the comparison results between our predicted characters and corresponding characters created by the artist to indicate our proposed model has a good performance on this task, and we also created new characters to show the “creativity” of our model.

CCS CONCEPTS

• Applied computing → Media arts.

KEYWORDS

calligraphic art, transformer, deep learning

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1 INTRODUCTION

Square word calligraphy is an art character system created by the artist Xu Bing [Xu [n.d.]]. He composes modified English alphabets into a square format that is similar to Chinese characters. Inspired by this, we want to study a deep learning model to learn this system. We treat the production of such calligraphic characters as an assembly task, the calligraphy as synthesis by transforming each English alphabet.

Assembly work has been studied and explored by many researchers. The assembly of parts is a complicated and laborious task. Results in the complicated structures of parts and the task of assembly involves exploring an extremely large solution space and reasoning the input part relationship for candidate assembly

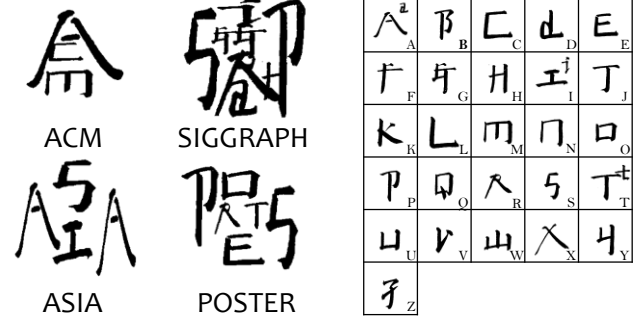


Figure 1: On the left results of “ACM SIGGRAPH ASIA POSTER” we generated, and on the right is an alphabet-calligraphy corresponding table created by artist Xu Bing.

proposals. [Huang et al. 2020] focus on the pose estimation sub-problem from the vision side involving geometric and relational reasoning over the input part geometry, and use an iterable graph neural network to gradually refine the pose estimation of the part from coarse to fine. [Li et al. 2020] learn a part-aware generative network, named as PAGENet proposed to model the 3D part structure in more detail so that improve the accuracy of the assembly target. We cannot directly use the previous assembly method that artwork assembly tasks often have a richer appearance and involve more complicated relationships between parts and between parts and the whole. The obvious difference is that there are often semantic parts in previous works. For example, the semantic parts of chairs include backrests, armrests, cushions, feet, etc. But there are no meaningful semantic parts in artwork assembly tasks, and parts do not often have a fixed position. It pays more attention to the coordination between parts and the whole, and the style is consistent.

To achieve this assembly task, we modified Transformer [Vaswani et al. 2017] with a random mask, We have conducted experiments and obtained superb predicted results referencing the characters created by the artist. Moreover, we create new characters by our proposed model as shown in Fig. 1.

2 APPROACH

2.1 Architecture based on Transformer

As shown in Fig. 2, the backbone of our model is based on transformer model. The input of the entire model is an English word $\chi \in X$. The output is a $k \times 4$ 4-DoF transformation of each alphabet in the word, including translation, scaling, width, height respectively (x, y, w, h) . We set the maximum length of words as k . The model consists of one encoder with 6 encoder layers and two different decoders with 6 decoder layers. Layers have the same network structure respectively in encoder and two decoders. A multi-head attention module is provided in each encoder layer, and the input

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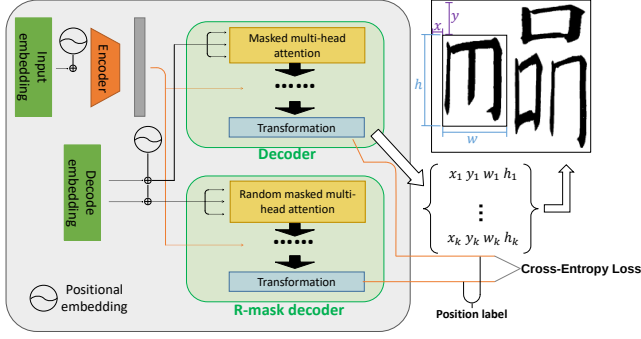


Figure 2: Architecture of our model.

of the entire encoder is a 512-dimensional word embedding code. A mask multi-head attention module and a multi-head attention module are provided in each decoder layer. The input of the entire decoder consists of two, one is the output coding from the encoder and the other is the embedding of alphabets that have already known (respectively from Sequential mask and random mask). Positional embedding is added after all input embedding. In the second decoder, we replaced the mask multi-head attention module with a random one (Using random mask could generate different output results to increase the diversity of results) with a mask rate β of each alphabet coding. To ease the training, we normalize all scaling and translation parameters into $[0, 1]$ based on an allowed range of scaling ratio $[[0.0, 2.0]]$ and translation $[[0, 512]]$, with the unit being pixel size). The actual values of scaling and translation parameters are recovered when being applied. We define the network as P and the corresponding ground-truth of input χ as $G(\chi)$. We employ the Cross-Entropy Loss to reduce the difference between the output transformation and the real transformation. Our loss $L = L_1 + L_2$ can be described as the following two parts:

$$L_1(G(\chi), P(\chi)) = - \sum_{\chi \in X} (G(\chi) \cdot \log P(\chi)) \quad (1)$$

$$L_2(G(\chi), P_{mask}(\chi)) = - \sum_{\chi \in X} (G(\chi) \cdot \log P_{mask}(\chi)) \quad (2)$$

where $P_{mask}(\chi)$ is the output from random mask decoder.

2.2 Data set and training

We collect 369 characters of Square Word Calligraphy by searching the published websites and manually label the position of each alphabet in each data for calculating their translation and scaling. Due to the small number of characters created by the artist, we apply data augmentation by replacing the alphabet characters in the image with others. We randomly select 300 data for augmentation and use them as the training set, and the remaining 69 data are used as our test set. During training, we define maximum length $k = 11$, mask rate $\beta = 0.5$. We use Adam optimizer to train our model and set learning rate $\eta = 0.0001$, $\beta = (0.5, 0.99)$, and the training epoch is 40.

3 RESULT

We show examples of predicted results, the corresponding ground truths as well as input English words in Fig. 3. Our designed model obtained characters that are similar to Chinese characters even it



Figure 3: Predicted examples. For each character, we show the predicted results in the middle, the corresponding ground truth (created by artist) on the upper left, and the input English below.

has a different structure with ground truth. To evaluate our results, we invited three Chinese students to classify them into three grades: outstanding, good, and poor, by referencing the ground truth and considering if they are similar to the format of Chinese characters. We obtained that 74% of our predicted results are “outstanding” or “good” for resembling Chinese characters on average.

4 CONCLUSION AND LIMITATION

In this paper, we propose a transformer-based model to reassemble English alphabets into a format of Chinese square characters. For the limitation, our current model does not perform well for the generation of longer English words. Moreover, the strokes of the generated calligraphy are single and lack artistic sense since our method is based on reassembly.

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